Catch-Rematched Hybrid Flare-Gated Transit: A Packet-Centered Composite Wormhole–Warp Architecture

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Abstract

This paper presents a standalone reduced-engineering reference model for a composite faster-than-light spacetime architecture under the conditional assumption that the required null-violating or effective source support can be supplied. The construction combines a hybrid flare-gated wormhole-support infrastructure with a catch-rematched throat-loaded transit packet. The central design move is a passenger/infrastructure separation: the throat complex supplies capacity, lapse support, radial stretch, flare geometry, shift containment, transition timing, and source-current management, while passenger safety is attached to a finite protected packet worldtube. The frozen candidate uses early-catch transition choreography, throat-gated shift, broadened mixed radial/angular capacity support, and moderate broad radial service stretch. In the reduced ADM screens, transition-profile choices control the dynamic/current burden measured by j_M and K, while capacity/support geometry controls the Hamiltonian/spatial-curvature burden measured by ρ_H and ${}^{(3)}R$. The confirmed candidate

$$C_0 = 20, \quad C_\perp = 5, \quad R_{\rm th} = 1.25, \quad w_{\rm th} = 0.12, \quad R_{\rm pass} = 0.35, \quad w_{\rm pass} = 0.08, \quad B_0 = 6, \quad w_B = 12$$

with transition parameters

$$x_{\rm catch} = 0.05, \quad x_\beta = 0.70, \quad x_q = 1.25, \quad w_{\rm catch} = 0.25, \quad w_\beta = 0.28, \quad w_q = 0.30, \quad p_\beta = 4$$

clears packet and support-edge causal checks in the reported reduced screens. Relative to the v01 geometry baseline, the candidate reduces the confirmed total source-demand proxy to about 0.162×, the geometry channel to about 0.274×, and the dynamic channel to about 0.0108× on the three-scenario confirmation grid. A conservative current-floor source screen reduces Lorentzian-sampled packet negative exposure at $V = 10, \lambda = 6, \tau = 0.2$ to about 0.042× the baseline. Robustness checks preserve the mixed-capacity design family under pressure-sensitive proxy choices, score-weighting changes, and grid variation. The paper's contribution is a concrete composite-design map: transition profiles manage current burden, mixed radial/angular capacity manages spatial-curvature burden, and packet safety is evaluated by the moving packet worldtube rather than by passive throat comfort.

1 Purpose and main result

Traversable wormholes, warp drives, Krasnikov-style route infrastructure, and ADM warp-shell models each isolate part of the faster-than-light engineering problem. Wormhole work identifies throat support, flare-out, tidal, and chronology constraints. Warp-drive work separates mobile shift, wall thickness, spatial capacity, and lapse/time-rate structure. Prepared-route models emphasize infrastructure and causality control. Controlled traversability work shows that source timing can make a channel traversable in special theoretical settings. The present paper combines these lessons into a single reduced composite architecture.

The result is a packet-centered active-rail geometry. The throat is treated as an active infrastructure plant. The passenger is represented by a finite coupling packet. The hard operating rule is

$$\operatorname{supp}(\beta^l) \subseteq \operatorname{supp}(A, T), \qquad x_{\text{catch}} < x_\beta < x_q, \tag{1}$$

with the operational sequence

$$\text{catch} \to \text{fade shift} \to \text{relax throat}. \tag{2}$$

The passenger diagnostic is the moving packet norm,

$$g_{\mu\nu}\dot{x}^\mu_{\text{pkt}}\dot{x}^\nu_{\text{pkt}} < 0, \tag{3}$$

with support-edge containment tracked separately by stationary and ADM diagnostics.

The frozen design is called the Catch-Rematched Hybrid Flare-Gated Transit v1. Its design doctrine is

$$\boxed{\text{early-catch transition choreography + broadened mixed radial/angular capacity support + moderate broad radia}} \tag{4}$$

The candidate parameters are listed in Table 1. The nearby $C_\perp = 3$ branch is recorded as a pressure-leaning operating variant; the balanced default uses $C_\perp = 5$.

Table 1: Frozen balanced default candidate. The transition layer comes from the v01 transition refinement; the capacity/support geometry comes from the v02 geometry and robustness screens.

Subsystem	Parameters
Transition timing	$x_{\text{catch}} = 0.05$, $x_\beta = 0.70$, $x_q = 1.25$
Transition widths	$w_{\text{catch}} = 0.25$, $w_\beta = 0.28$, $w_q = 0.30$
Edge shift shaping	$p_\beta = 4$
Radial capacity	$C_0 = 20$
Angular capacity	$C_\perp = 5$
Throat support	$R_{\text{th}} = 1.25$, $w_{\text{th}} = 0.12$
Packet support	$R_{\text{pass}} = 0.35$, $w_{\text{pass}} = 0.08$
Radial service stretch	$B_0 = 6$, $w_B = 12$
Service flare posture	R open or delayed through packet service

2 Literature setting and specific gap

The literature supplies the physical constraint map. Morris and Thorne establish the traversable throat as a flared areal-radius minimum requiring exotic radial null support and low tidal forces for traversability [1]. Morris, Thorne, and Yurtsever connect wormhole engineering to chronology control when separated mouths are used [2]. Ford and Roman show that quantum inequalities impose severe magnitude-duration and length-scale pressure on macroscopic wormhole support, with negative energy tending toward thin regions or Planck-scale structures in the static cases they analyze [3]. Fewster and Roman sharpen the null-contracted sampled-stress issue for small-exotic-matter proposals [4]. Dynamic-throat work keeps local null expansions and local energy-condition pressure central [5].

Warp-drive work supplies the ADM and infrastructure vocabulary. Alcubierre introduced a superluminal warp geometry with exotic source requirements [6]. Pfenning and Ford connected the Alcubierre wall to quantum-inequality severity and extreme wall thinness [7]. Van den Broeck showed that capacity engineering can change energy accounting by combining compact exterior structure with large interior volume [8]. Natario showed that expansion/contraction is not the essential diagnostic of a warp-drive geometry, emphasizing the role of the shift structure [9]. Everett and Roman's Krasnikov-tube analysis emphasizes prepared infrastructure, causality, and negative-energy layers in route-based superluminal travel [10]. Bobrick and Martire provide a broad ADM decomposition of warp-drive spacetimes into lapse, shift, spatial geometry, capacity, and time-rate roles [11]. Recent positive-energy and subluminal warp-shell work reinforces the importance of matter-shell design and shift-vector engineering [12, 13]. Gao, Jafferis, and Wall show that traversability can arise from a controlled source-timing mechanism in a holographic setting [14].

The gap addressed here is design-specific. The literature gives the principles: shell thickness matters, capacity matters, ADM role separation matters, prepared infrastructure matters, and source timing matters. The present work supplies a concrete composite mapping:

$$\text{transition profiles} \longrightarrow j_M,\ K, \tag{5}$$

$$\text{capacity/support geometry} \longrightarrow \rho_H,\ {}^{(3)}R. \tag{6}$$

It also supplies a new capacity rule for this composite architecture: radial service capacity and angular capacity should be separated. The angular sector is a spatial-curvature burden reducer in the packet-service geometry. In the confirmed screen, the no-angular comparison $C_\perp = 1$ leaves the geometry channel near the v01 baseline, while $C_\perp = 5$ reduces the confirmed geometry cost to about $0.274\times$ and total cost to about $0.162\times$ relative to the v01 geometry baseline.

3 Prior ingredients used by the composite model

The composite design draws from three reduced design lines, all recast here as standalone ingredients.

3.1 Hybrid flare-gated throat infrastructure

The wormhole infrastructure uses a spherically symmetric metric family

$$ds^2 = -N(l,t)^2 \mathrm{d}t^2 + B(l,t)^2 \mathrm{d}l^2 + R(l,t)^2 \mathrm{d}\Omega^2. \tag{7}$$

The infrastructure assignments are

$$\begin{aligned} B(l,t):&\ \text{radial support dilution and service-coordinate stretch,}\\ R(l,t):&\ \text{flare/access-state and angular geometry control,}\\ N(l,t):&\ \text{timing, shoulder matching, and lapse margin.} \end{aligned}$$

The wormhole-only reference model uses a phase sequence

$$B \text{ prestretch} \to R \text{ flare opening} \to \text{quiet access hold} \to R \text{ closure} \to \text{repayment/shoulder matching} \to B \text{ reset}. \tag{8}$$

Its reduced source architecture separates NMC-like support, infrastructure repayment, and shoulder/matching components. In the wormhole-only closure, full-cycle flux-completed observer-family ratios clear unity for the monitored core, access, support, support-ring, and shoulder ledgers [15].

In the composite design, this wormhole infrastructure becomes active plant. The passenger criterion moves to a protected packet. The throat supports capacity, lapse, shift containment, release timing, and source-current management.

3.2 Catch-rematched throat-loaded packet

The original combined geometry introduced a throat-loaded ADM radial-axis form,

$$ds^2 = -T^2\mathrm{d}t^2 + A^2\left(\mathrm{d}l + \beta^l\mathrm{d}t\right)^2 + A^2 r(l)^2\mathrm{d}\Omega^2. \tag{9}$$

The capacity and lapse support are paired,

$$A = \exp(qW\ln C_0), \qquad T = \exp(qW\ln(\lambda C_0)), \tag{10}$$

so that

$$\frac{T}{A} = \lambda^{qW}. \tag{11}$$

The shift is localized in the packet and gated by throat support,

$$\beta^l(l,t) = -U(t)E(X(t))W(l)S(l - X(t)). \tag{12}$$

The support-inclusion rule $\mathrm{supp}(\beta^l) \subseteq \mathrm{supp}(A,T)$ follows from the gated-shift screens: the independent passenger-shift branch generated lapse-shift failures, while the throat-gated branch cleared the tested traversal and exit configurations [16].

3.3 Engineering screening framework

The wormhole-support engineering framework decomposes a proposed spacetime plant into support, access, repayment, buffer, matching, transition, and source-realism functions, then assigns diagnostic gates to each function [17]. The composite work applies that framework to a combined wormhole-warp system. The source-support assumption is held fixed; the work reported here maps the engineering consequences of that assumption.

4 Composite ansatz

The final composite ansatz inserts the catch-rematched packet into the hybrid flare-gated throat controls:

$$ds^2 = -\alpha(l,t)^2\mathrm{d}t^2 + \gamma_{ll}(l,t)\left(\mathrm{d}l + \beta^l(l,t)\mathrm{d}t\right)^2 + \gamma_{\Omega\Omega}(l,t)\mathrm{d}\Omega^2. \tag{13}$$

The fields are

$$\alpha(l,t) = N_{\mathrm{v1}}(l,t)T_{\mathrm{pkt}}(l,t), \tag{14}$$

$$\gamma_{ll}(l,t) = B_{\mathrm{v1}}(l,t)^2 A_{\parallel}(l,t)^2, \tag{15}$$

$$\gamma_{\Omega\Omega}(l,t) = R_{\mathrm{v1}}(l,t)^2 A_{\perp}(l,t)^2, \tag{16}$$

$$\beta^l(l,t) = -\dot{X}_{\mathrm{coord}}(t)E(X(t))W(l)^{p_\beta}S(l - X(t)). \tag{17}$$

The packet uses mixed radial/angular capacity:

$$A_{\parallel} = \exp(qW\ln C_0), \tag{18}$$

$$A_{\perp} = \exp(qW\ln C_{\perp}), \tag{19}$$

$$T_{\mathrm{pkt}} = \exp(qW\ln(\lambda C_0)). \tag{20}$$

The coordinate speed is service-stretch aware,

$$\dot{X}_{\mathrm{coord}} = \frac{U}{B_{\mathrm{v1}}(X)}. \tag{21}$$

This conversion is an engineering requirement of the composite model: B_{v1} stretches radial proper length, so a proper-radial service speed must be translated into coordinate motion before it is used in the ADM shift.

The protected packet region is sampled as

$$|l - X(t)| \le r_{\mathrm{pkt}}. \tag{22}$$

The stationary infrastructure diagnostic is

$$g_{tt} = -\alpha^2 + \gamma_{ll}(\beta^l)^2. \tag{23}$$

The moving packet diagnostic is

$$\mathcal{N}_{\mathrm{pkt}} = -\alpha^2 + \gamma_{ll}\left(\dot{X}_{\mathrm{coord}} + \beta^l\right)^2. \tag{24}$$

The packet clears the causal check when $\mathcal{N}_{\mathrm{pkt}} < 0$ across the sampled packet region.

5 Diagnostics and cost functions

The reduced ADM screens use the spatial metric γ_{ij}, lapse α, shift β^i, and extrinsic curvature

$$K_{ij} = \frac{D_i\beta_j + D_j\beta_i - \partial_t\gamma_{ij}}{2\alpha}. \tag{25}$$

The Hamiltonian source-demand proxy is

$$\rho_H = \frac{{}^{(3)}R + K^2 - K_{ij}K^{ij}}{16\pi}, \tag{26}$$

and the momentum/current source-demand proxy is

$$j^i_M = \frac{D_j(K^{ij} - \gamma^{ij}K)}{8\pi}. \tag{27}$$

The reported norm is $|j_M| = (\gamma_{ij} j^i_M j^j_M)^{1/2}$.

The screens use two burden channels:

$$J_{\mathrm{geom}} = \max|\rho_H| + 0.05\max|{}^{(3)}R|, \tag{28}$$

$$J_{\mathrm{dyn}} = \max|j_M|_{\mathrm{pkt}} + \max|j_M|_{\mathrm{edge}} + 0.02\max|K|_{\mathrm{pkt}}, \tag{29}$$

$$J_{\mathrm{total}} = J_{\mathrm{geom}} + J_{\mathrm{dyn}}. \tag{30}$$

The exact weights are screening choices; robustness checks below confirm the family under alternative weights. The source-exposure proxy is a conservative current floor,

$$Tkk_{\mathrm{floor}} = \rho_H - 2|j_M|. \tag{31}$$

A pressure-sensitive bracket was also checked using $\rho_H + \eta\rho_H - 2|j_M|$ with $\eta \in \{-1, 0, +1\}$ and a worst-sign bracket $\rho_H - |\rho_H| - 2|j_M|$.

6 Transition refinement: v01

The transition refinement fixed the support geometry and optimized catch/shift/relax profiles. The best confirmed transition candidate is

$$x_{\text{catch}} = 0.05, \quad x_\beta = 0.70, \quad x_q = 1.25, \qquad w_{\text{catch}} = 0.25, \quad w_\beta = 0.28, \quad w_q = 0.30, \qquad p_\beta = 4. \tag{32}$$

The result is shown in Table 2. The main transition lesson is that early catch and medium-width profiles reduce dynamic/current burden while deepening the packet's timelike margin. The smoothest tested transition spread too far across the service phases and failed packet clearance. Strong edge shaping with W^4 reduced j_M and K relative to lower p_β in this family.

Table 2: Transition-layer refinement at the stressed $V = 10, \lambda = 6$ screen. Values are relative to the baseline transition where noted.

Metric	Baseline transition	Early-catch medium $p_\beta = 4$
Packet max norm	-0.194	-0.704
Release max packet norm	-0.194	-1.819
Edge max g_{tt}	-1.187	-1.350
Max packet j_M proxy	1.000	0.603
Max packet K proxy	1.000	0.601
Total source-cost proxy	1.000	0.939

The transition result identifies the first burden channel:

$$\boxed{\text{catch/shift/relax profile design controls } j_M \text{ and } K.} \tag{33}$$

7 Geometry refinement: mixed capacity support

With the v01 transition layer fixed, the geometry sweep varied C_0, $C_\perp$, R_{th}, w_{th}, R_{pass}, w_{pass}, B_0, and w_B. The confirmation screen used three scenarios,

$$(V, \lambda) = (5, 5.75), \quad (10, 6), \quad (10, 11.5), \tag{34}$$

and service positions

$$X \in \{0.05, 0.35, 0.70, 1.00, 1.25\}. \tag{35}$$

The frozen default candidate is $C_0 = 20, C_\perp = 5, R_{\text{th}} = 1.25, w_{\text{th}} = 0.12, R_{\text{pass}} = 0.35, w_{\text{pass}} = 0.08, B_0 = 6, w_B = 12$.

Table 3 gives the key confirmed candidates. The best balanced default reduces total cost to $0.162\times$ the v01 geometry baseline, with packet and edge failures equal to zero.

The geometry result identifies the second burden channel:

$$\boxed{\text{capacity/support geometry controls } \rho_H \text{ and } {}^{(3)}R.} \tag{36}$$

It also identifies the design role of $C_\perp$: angular capacity reduces spatial-curvature concentration in the packet-service geometry. This is the specific new design result that was not fixed by the general literature principles alone.

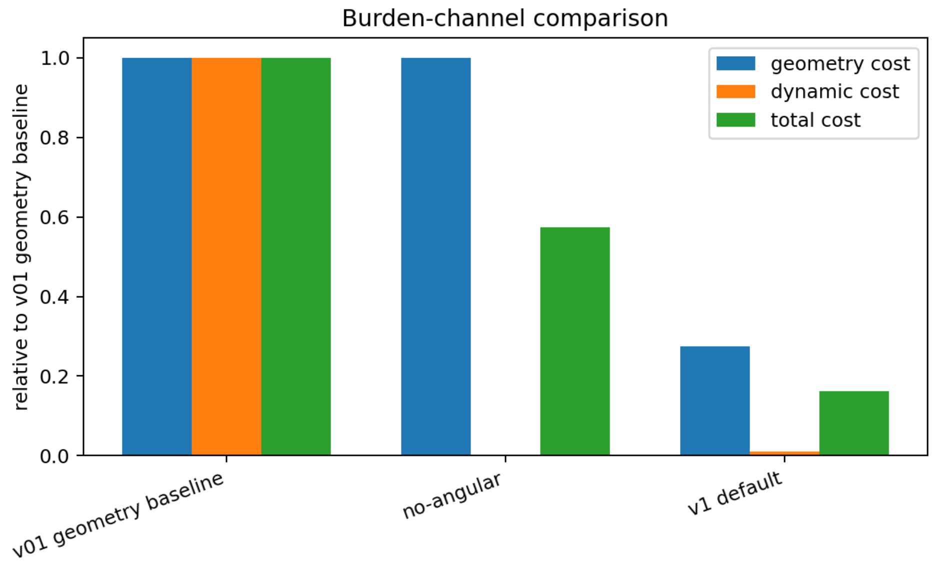

Figure 1: Burden-channel comparison on the three-scenario confirmation grid. The v1 candidate reduces the geometry channel and dynamic channel together; the no-angular comparison strongly reduces dynamic burden while retaining the geometry burden.

Table 3: Geometry confirmation grid. All listed candidates clear packet and edge checks. Costs are relative to the v01 geometry baseline.

Design	J_{total}	J_{geom}	J_{dyn}	max packet norm	max edge g_{tt}
$v01$ geometry baseline	1.000	1.000	1.000	−0.696	−128.5
No-angular $C_\perp = 1$	0.574	1.000	0.002	−0.762	−3.001
$C_\perp = 3$ branch	0.174	0.298	0.00776	−0.762	−3.001
$C_\perp = 5$ balanced default	0.162	0.274	0.0108	−0.762	−3.001
Reserve $C_0 = 100, C_\perp = 5$	0.174	0.278	0.0341	−0.766	−4.356

8 Quantum-source proxy screen

The pre-freeze source screen evaluates the conservative current-floor proxy in Eq. (31). At the stressed $V = 10, \lambda = 6$ case, the default candidate reduces the Lorentzian-sampled packet floor at $\tau = 0.2$ from -2.0645×10^{-2} to -8.704×10^{-4}, about $0.042\times$ the v01 baseline. It also reduces the packet negative-volume proxy to about $0.092\times$, packet j_M to about $0.011\times$, and packet ρ_H and $^{(3)}R$ to about $0.305\times$ the baseline. Table 4 summarizes the result.

This screen clarifies the no-angular trade. The no-angular comparison is current-clean and geometry-heavy. The selected $C_\perp = 5$ candidate is balanced: it keeps current exposure small while reducing the Hamiltonian/spatial-curvature burden that a source model must carry. The source screen therefore supports the freeze as a mixed-capacity design rather than a pure current-minimization design.

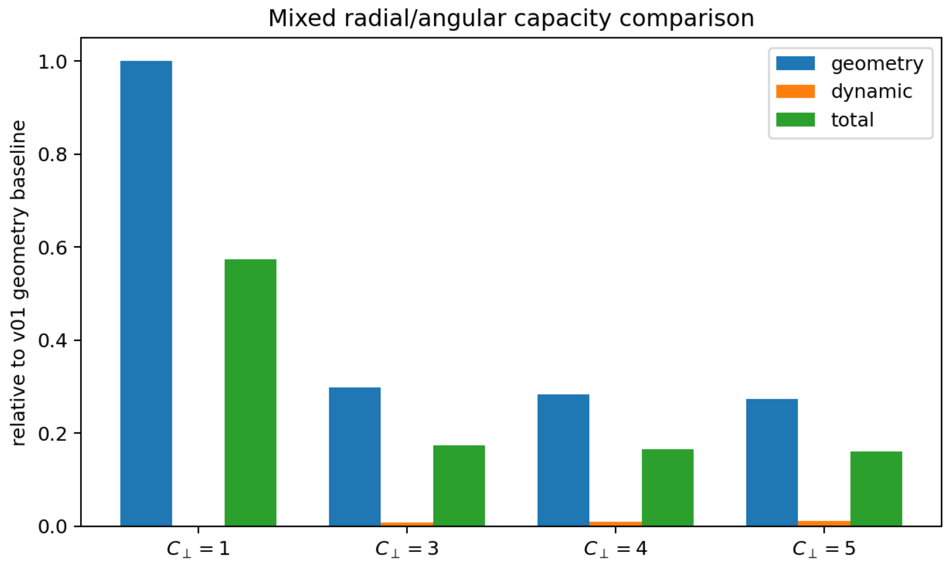

Figure 2: Mixed radial/angular capacity comparison. The $C_\perp = 1$ no-angular candidate is dynamically quiet and geometry-heavy. Moderate angular capacity cuts the geometry burden while preserving packet and support-edge clearance.

Table 4: Conservative source-exposure proxy at $V = 10, \lambda = 6$. Relative values are against the v01 geometry baseline.

Design	sampled packet floor	packet neg. volume	packet j_M	packet ρ_H
$v01$ baseline	1.000	1.000	1.000	1.000
No-angular $C_\perp = 1$	positive floor	0.291	0.00134	1.000
Balanced default $C_\perp = 5$	0.042	0.092	0.0111	0.305
Reserve $C_0 = 100, C_\perp = 5$	0.295	0.130	0.0462	0.317

9 Robustness checks before freeze

Three low-cost robustness checks were run before freezing the candidate.

First, pressure-sensitive brackets were evaluated with

$$Tkk_\eta = \rho_H + \eta\rho_H - 2|j_M|, \qquad \eta \in \{-1, 0, +1\}, \tag{37}$$

and with $Tkk_{\text{worst}} = \rho_H - |\rho_H| - 2|j_M|$. At 61×25 grid resolution in the stressed case, the $C_\perp = 5$ default keeps total cost at $0.119\times$, geometry cost at $0.158\times$, dynamic cost at $0.020\times$, floor exposure at $0.338\times$, and pressure-worst exposure at $0.338\times$ relative to the v01 baseline. The $C_\perp = 3$ branch gives lower pressure-sensitive negative exposure and a slightly higher total cost. The useful design band is therefore $C_\perp \simeq 3$–5, with $C_\perp = 5$ as the balanced default.

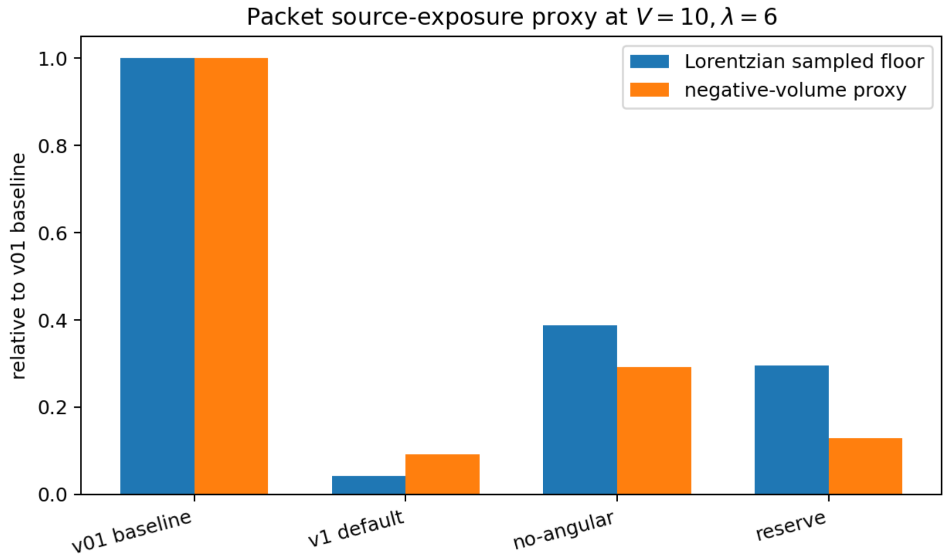

Figure 3: Packet source-exposure proxy at $V = 10, \lambda = 6$. The balanced default reduces sampled negative packet exposure more strongly than the total ADM cost reduction alone would suggest, because it suppresses the current channel that dominates the negative floor.

Second, score-weighting robustness tested

$$
\begin{aligned}
J_1 &= J_{\mathrm{geom}} + J_{\mathrm{dyn}}, \\
J_2 &= J_{\mathrm{geom}} + 0.1 J_{\mathrm{dyn}}, \\
J_3 &= 0.1 J_{\mathrm{geom}} + J_{\mathrm{dyn}}, \\
J_4 &= \max(\text{normalized metrics}).
\end{aligned}
$$

The top family remained mixed-capacity with $C_\perp = 3$–5, $C_0 = 20$–35, $B_0 \simeq 6$, and broad w_B. This establishes the family rather than a single scoring artifact.

Third, grid sensitivity was checked at 25×11, 41×17, and 61×25. The ordering persists:

$$
\text{mixed-capacity default} < \text{no-angular comparison} < \text{v01 baseline} \tag{38}
$$

for total cost. Packet and edge failures remain zero for the default candidate.

Table 5: Grid-sensitivity total-cost ratios.

Design	25×11	41×17	61×25
Balanced default $C_\perp = 5$	0.0608	0.1617	0.1194
No-angular $C_\perp = 1$	0.1707	0.5738	0.7213
$v01$ baseline	1.000	1.000	1.000

The robustness result supports the v1 freeze. The default candidate is the balanced mixed-capacity design. The $C_\perp = 3$ branch is a nearby pressure-leaning operating variant.

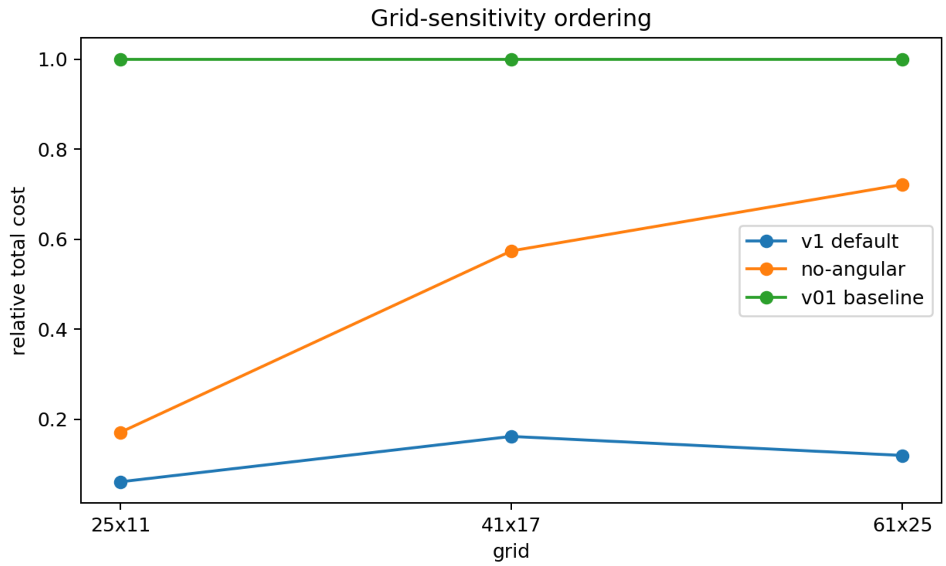

Figure 4: Grid-sensitivity ordering. The ratios move with resolution, while the design ordering remains stable.

10 Engineering interpretation

The composite system shifts the engineering problem from a globally comfortable throat to an active packet-service apparatus. The throat complex carries support machinery. The packet carries the passenger criterion. The diagnostics follow the object being protected:

$$\begin{aligned} g_{tt} < 0 &: \text{ stationary infrastructure accessibility,} \\ \mathcal{N}_{\text{pkt}} < 0 &: \text{ moving passenger/coupling packet safety.} \end{aligned}$$

This distinction explains why low-lapse reduced tests can trigger stationary g_{tt} while the packet remains timelike under correct choreography.

The v1 design map is:

1. B is a radial service-coordinate and support-smoothing actuator.
2. R is a service flare and angular geometry actuator; the active-rail posture keeps R open or delayed through service and relaxes it smoothly.
3. N supplies timing and shoulder/lapse margin.
4. W^{p_β} edge shaping cleans support edges after correct catch timing.
5. Early catch reduces dynamic current demand and deepens packet timelike margin.
6. Mixed radial/angular capacity reduces the spatial-curvature burden.

The strongest new capacity statement is

$$\boxed{\text{minimum useful radial capacity plus moderate angular capacity is favored over purely radial capacity.}} \tag{39}$$

The data behind this statement are the $C_\perp = 1$ and $C_\perp = 5$ comparison: the no-angular candidate is current-clean and geometry-heavy; the balanced default reduces both packet exposure and geometry burden.

11 Relationship to the broader field

The field already expects thin walls, compact high-gradient support, and uncontrolled negative-energy layers to be expensive. The present results supply a more specific engineering answer for a composite wormhole-warp design. Table 6 summarizes the mapping.

Table 6: Literature lesson and composite-design clarification.

Literature lesson	Composite-design clarification
Morris–Thorne throat flare-out and exotic support	The throat is active infrastructure; the protected object is the packet worldtube.
Ford–Roman and Pfenning–Ford thin-layer severity	Broadened support and mixed capacity reduce reduced ADM source-demand and packet source-exposure proxies.
Van den Broeck capacity engineering	Capacity should be split into radial service capacity and angular curvature-relief capacity.
Everett–Roman prepared infrastructure	The catch-rematched throat complex is a local prepared-route service apparatus with explicit catch/shift/relax order.
Bobrick–Martire ADM role separation	The sweep maps transition profiles to j_M, K and capacity/support geometry to $\rho_H, {}^{(3)}R$.
Gao–Jafferis–Wall source timing	The composite design turns timing into a concrete service basin: catch, fade shift, relax throat.

The paper therefore advances the engineering understanding under the source-availability assumption. It identifies which knobs move which burden and gives a frozen reduced candidate for the next source-construction and constraint-quality phases.

12 Scope and remaining gates

The reported work is a prescribed-geometry reduced engineering screen. Its claims are bounded by the diagnostics defined in this paper. The work supplies a source-demand and source-exposure target. The next gates are:

1. pressure-complete source decomposition $T_{\mu\nu}k^\mu k^\nu$ for the frozen candidate;
2. source-channel construction for support, transition-current, angular-capacity, shoulder/matching, and packet-coupling components;
3. full constraint-quality initial data beyond the reduced ADM proxy;
4. evolution, stability, ringing, and repeated-cycle accumulation;
5. chronology and network-operation constraints.

These are follow-on gates. The frozen v1 contribution is the reduced composite architecture and its burden-channel map.

13 Reproducibility package

The accompanying bundle includes the LaTeX source, compiled PDF, selected data tables, scripts, figures, source notes, and checksum manifest. The main scripts are:

- `scripts/adm_3p1_viability_v3_baware.py`: reduced ADM viability diagnostics for the exact-v1/catch-rematched composite;
- `scripts/run_transition_refinement_fast.py`: v01 transition-layer refinement;
- `scripts/run_v02_confirmation_narrow.py`: geometry confirmation sweep;
- `scripts/run_v02_quantum_source_screen.py`: conservative current-floor source screen;
- `scripts/run_candidate_robustness_checks.py`: pressure, scoring, and grid robustness checks;
- `scripts/build_paper_figures.py`: figure generation from bundled CSV data.

The included data are compact summary tables rather than full per-slice tensor archives. The scripts regenerate the reported tables and figures from the selected summaries.

14 Conclusion

This paper freezes a packet-centered composite wormhole-warp engineering reference model under the assumption that the required source support can be supplied. The model uses hybrid flare-gated throat infrastructure as active rail plant and a catch-rematched throat-loaded packet as the passenger/coupling object. The operating sequence is catch, fade shift, relax throat. The support-inclusion rule is $\mathrm{supp}(\beta^l) \subseteq \mathrm{supp}(A, T)$. The passenger criterion is the moving packet norm.

The main design result is burden-channel separation. Transition profiles control j_M and K; capacity/support geometry controls ρ_H and ${}^{(3)}R$. The v01 transition refinement identifies early catch, medium transition widths, and W^4 edge shaping. The v02 geometry refinement identifies broadened mixed radial/angular capacity support with moderate broad B stretch. The source screen and robustness checks support a frozen balanced default with $C_0 = 20$, $C_\perp = 5$, $R_{\mathrm{th}} = 1.25$, $w_{\mathrm{th}} = 0.12$, $B_0 = 6$, and $w_B = 12$, plus a nearby pressure-leaning $C_\perp = 3$ branch.

The work clarifies what the general literature leaves at the principle level. It turns wall-thickness, capacity, ADM role separation, prepared infrastructure, and source timing into a concrete composite-design map. The resulting v1 model is ready for source-channel construction, pressure-complete null-stress evaluation, and higher-fidelity constraint-quality studies.

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